
Fire-Guard

Persona & Use Scenarios

Persona & Hypothesis Validation Plan

TEAM: KGE

계명대 정원택

계명대 안현일

계명대 김관우

계명대 정현우

• Background & Problem

01

Context

- Large complex facilities and underground parking lots have many blind spots due to their vast and complex structures.

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- Surveillance relies solely on CCTV during unmanned hours like night shifts or weekends.

02

Problem

- **Invisibility:** Smoke blocks CCTV visibility, making it impossible to detect human presence.

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- **Wasted Golden Time:** Reliance on panicked reports or manually checking hundreds of screens delays initial response significantly.

● Target Persona

Profile: Charles Kim (45) / Safety Team Manager at Large Complex Facility

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Responsibilities :

Monitoring hundreds of CCTV feeds and managing emergencies.

Directing evacuation broadcasts, reporting to fire departments, and ordering initial suppression.

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Characteristics :

Constantly anxious about accidents occurring in blind spots.

Exhausted by false alarms; demands reliable and intuitive information.

● Pain Points

1. Location Confusion :

Alarms ring, but pinpointing the exact floor and corner takes too long.

2. Uncertainty of Occupancy :

Impossible to verify if people are trapped or evacuated during smoke or blackouts.

3. Monitoring Overload:

Nearly impossible for a small team to monitor countless CCTVs 24/7 without error.

• Solution Concept

- **Core Idea:** Fire-Guard Autonomous Patrol Robot

- **Key Features :**

1. **Thermal + RGB Fusion** : Accurately distinguishes fire sources and humans even in darkness and smoke.
2. **Coordinate-based Mapping** : Rather than simply displaying a video, it is possible to determine the robot ' s current location within the map.
3. **Active Surveillance** : Robots patrol to cover blind spots that fixed CCTVs miss.

• Key Hypotheses

01 Hypothesis 1 (Operational Efficiency): Displaying coordinates on a map will reduce situational awareness and dispatch time by over 50%.	02 Hypothesis 2 (Detection Reliability): Using thermal cameras will enable occupant identification with over 90% accuracy, even in smoke-filled environments.	03 Hypothesis 3 (Response Effectiveness): Providing evacuee location data to rescue teams in advance will significantly increase the success rate of rescue operations.
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• System Architecture

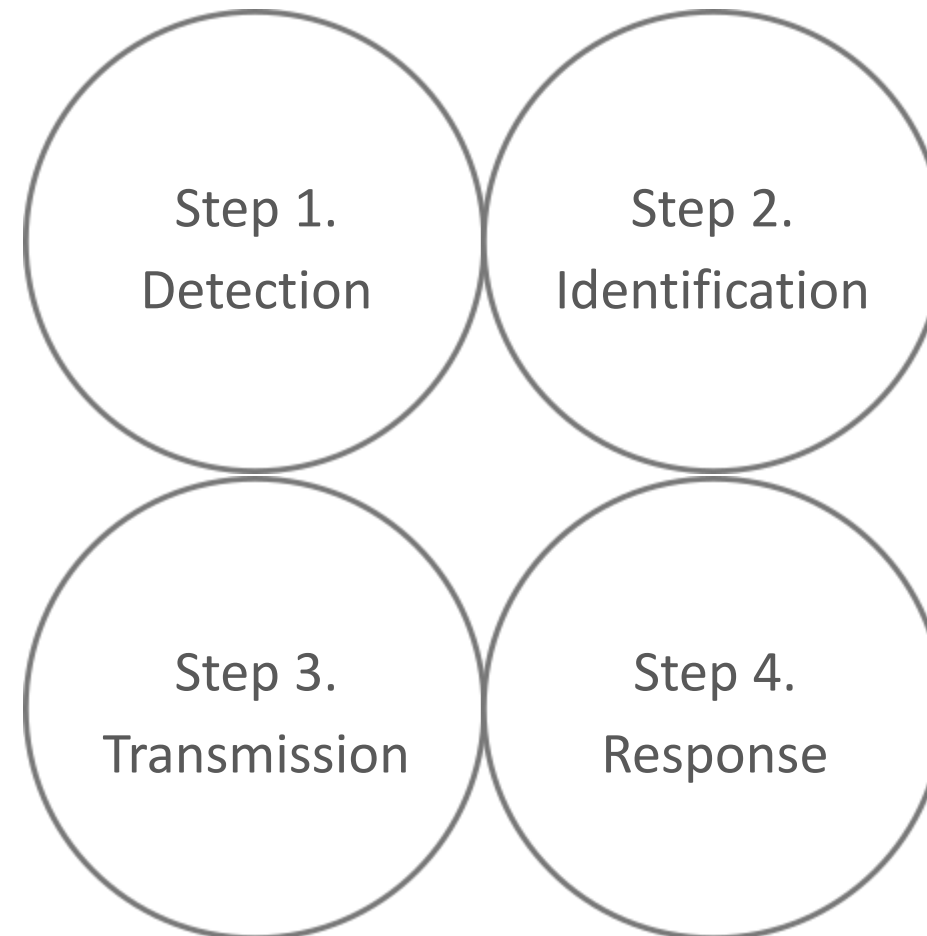
Hardware	- Mobility: LIMO-based autonomous robot (using ROS). - Sensors: LiDAR + Thermal Camera + RGB Camera.
Software	- SLAM: Real-time mapping and localization. - AI Vision: Algorithms for Fire (flame/heat) and Object (human) detection. - Web Control UI: Visualization of robot location and detection data on the control monitor.
Network	Real-time data transmission via Wi-Fi/LTE.

• Use Case Scenario

01. Situation



02. Step



• Value & Vision

01. Expected Value

- **Securing Golden Time:** Maximizing rescue chances by minimizing location verification time.
- **Eliminating Blind Spots:** 24/7 seamless surveillance covering areas fixed CCTVs miss.



02. Vision

- **"We are not replacing humans, but becoming the manager's 'eyes', acting as a safety partner guarding the most dangerous areas."**